

Ronnie Mondal

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PROFESSIONAL SUMMARY

- PhD strategic program manager and higher education leader and a proven track record of operationalizing complex, multi-million-dollar academic and administrative initiatives in decentralized research environments.
- Proven institutional advisor and consensus builder with extensive experience serving as a primary trusted advisor to university executives, leading high-stakes cross-functional committees, and managing diverse stakeholder coalitions.
- Data-driven operations expert combining elite data analytics experience, including a IPO exit, with advanced ML expertise to optimize institutional funding structures, build automated tracking workflows, and deliver executive-level strategic blueprints.

EDUCATION

Virginia Tech	Blacksburg, VA USA
<u>Doctor of Philosophy in Chemistry</u> , Dissertation : Modeling the Role of Water in Protein Structure and Function	2021 – 2026
Graduate Certificate: <u>Science, Technology and Engineering in Policy</u>	2022 – 2023
Graduate Certificate: <u>Politics and Policy Studies of Science and Technology</u>	2022 – 2024
Graduate Certificate: <u>Future Professoriate</u>	2021 – 2024
Indian Institute of Technology Bombay	Mumbai, MH India
Bachelor of Science in Chemistry (Hons.), Minor in Biosciences and Bioengineering	2016 - 2020

WORK EXPERIENCE

Research Proposal Development Specialist III, Tulane University	Aug 2026 – present
• Assisting faculty across disciplines, with a primary focus on the School of Science and Engineering, with the preparation and submission of competitive federal and foundation grant proposals for research funding..	
President, Virginia Tech Graduate and Professional Student Senate (VT GPSS)	2024 – 2025
• Elected to lead a 70+ member shared governance body representing 7K+ graduate and professional students across Virginia Tech campuses, exercised fiscal authority over \$120K annual operating budget and strategically allocated \$175K in auxiliary funding while coordinating with senior university leadership on institutional priorities. • Chaired cross-functional task forces with executive university administration, Title IX, HR, and graduate students to help establish the Office of Sexual Violence Prevention and Education, defining mandate, operating model, and implementation workflows for graduate-student-facing policy execution. • Engineered an institutional centralized grievance and escalation workflow to mitigate systemic risk; mobilized consensus across conflicting university divisions and delivered university policy blueprints adopted by senior administration. • Directed the University Mission Initiative Committee tasked with expanding graduate enrollment by 2,300 students; engineered predictive models and structural blueprints for tuition, fees, and stipend adjustments to optimize a \$40M funding matrix. • After end of term, chaired the Commission for Graduate and Professional Student Affairs (2025-2026), steering institutional committees on university policy and managing policy reconciliation processes across diverse campuses and campus divisions.	
Business Analyst, Swiggy (Bundl Technologies Pvt Ltd, Bengaluru, India)	2020-2021
• Scaled proprietary brand portfolios across hyper-competitive regional markets by leveraging predictive analytics and user-experience data, driving a 15% increase in engagement and a 22% surge in top-line growth. • Designed and coded interactive data-visualization dashboards for real-time campaign performance tracking; reduced reporting latencies and provided decision-support models for executive leadership. • Enabled higher market penetration and competitive positioning of private brands by analyzing complex datasets to identify high-impact products and optimize brand visibility across multiple cities, informing prioritization and go-to-market execution.	
Director of Advocacy, VT GPSS	2023-2024
Writer/Editor, Plot Politics	2021-2024
Logistics and Communication Head, TEDx IIT Bombay	2019
Editor-in-Chief, Spandan – The Vaani Magazine, IIT Bombay	2018-2019
Vice Contingent Leader, Inter-IIT Cultural Meet (Winning Team)	2018

TECHNICAL SKILLS

- **Policy and Governance:** Policy research, writing, editing, governance process coordination, stakeholder engagement, committee coordination, policy implementation support, compliance-minded documentation, training material development.
- **Strategic & Institutional Leadership:** Shared governance, cross-functional alignment, policy architecture, conflict mediation, risk mitigation, instructional design, project management, deadline management, administrative coordination, problem-solving.
- **Data Analytics & Digital Systems:** SQL, Python for data science, Tableau, Power BI, Banner SIS, Content Management Systems (CMS), dashboard architecture.

ACHIEVEMENTS

- Aspire! Award for Courageous Leadership 2025, Virginia Tech Student Affairs
- Nominated by Virginia Tech for the AAAS/Science Program for Excellence in Science, '26 – present.
- Policy and governance work covered by WVTF Public Radio, WDBJ7, ExperienceVT Blog, VT Governance Corner.
- Member, Academy for Outstanding Graduate Pedagogy, Virginia Tech, '25 – present.
- Best Poster Award Runner-Up, Southeastern Theoretical Chemistry Association Meeting '24.
- Graduate Teaching Award, Virginia Tech Department of Chemistry '23

COURSES AND CERTIFICATIONS

- **McKinsey Forward Program, McKinsey & Co** (Analytics, Problem-solving, and Leadership), Dec '25.
- Introduction to Computational Antibody Engineering, Schrödinger, Jun '24.
- Visualizing Science with PyMOL3, Schrödinger, Jun '24.
- Comprehensive credentials, verifications, and an extended repository of certifications are hosted on LinkedIn.

RESEARCH EXPERIENCE

PhD: Pharmaceutical Ion Channel Modeling for Drug Development

- Developed a Python-based simulation framework to study sodium ion transport in voltage-gated channels, informing rational drug design decisions in pain management therapies, using first-of-a-kind Continuous Time Random Walk (stochastic) model.
- Developed predictive models to analyze drug-relevant ion channels and biomaterials in early-stage drug development, applying machine learning, stochastic and Markovian modeling to generate decision-support signals for drug development research.

PhD: Collagen Modeling for Biophysics Applications

- Performed polarizable molecular dynamics simulations on collagen and mutants to evaluate biomolecular stability under varying conditions, supporting design strategies for tissue engineering and regenerative medicine applications.
- Built and released open-source software HeliXplore written in Python, to analyze biomolecular structures relevant to aging processes and biomedical materials design, enabling reproducible analysis and cross-functional research collaboration.

PhD: Calcium – Chitosan Interaction project

Collaboration with Dr Patricia Dove, Department of Geosciences, Virginia Tech

- Correlated sulfation levels in chitosan derivatives with calcite nucleation behavior using polarizable MD, revealing molecular mechanisms of polysaccharide-crystal interactions.

PhD: Polyacrylate – Protein Binding project – Featured in VT News

Collaboration with Dr Adrian Figg, Dept of Chemistry, & Dr Sanket Deshmukh, Dept of Chemical Engineering, Virginia Tech

- Uncovered polyacrylate-protein interactions at the residue level for functionalized soft materials using polarizable MD.
- Discovered composition dependence and sequence independence for polyacrylate-protein binding using Monte Carlo methods.

Undergraduate Intern for Borane Chemistry, Purdue University

2019

JOURNAL AND SOFTWARE PUBLICATIONS

5. HeliXplore: A Python package for analyzing multi-strand helix deformations. **R. Mondal** and V. Vaissier Welborn, Software Release on Zenodo, 2026, DOI: [10.5281/zenodo.18331481](https://doi.org/10.5281/zenodo.18331481)
4. Using AMOEBA to Uncover Deformation and Hydration Anisotropy in Collagen Mimetic Peptides. **R. Mondal** and V. Vaissier Welborn, Journal of Physical Chemistry B, 2026, DOI: [10.1021/acs.jpcb.6c02718](https://doi.org/10.1021/acs.jpcb.6c02718)
3. Tuning polyacrylate compositions to recognize and modulate fluorescent proteins. D.C. Gomez, S. Seth, **R. Mondal**, et al., Angew Chem Intl Ed, 2025, DOI: [10.1002/anie.202520032](https://doi.org/10.1002/anie.202520032). (*labeled Very Important Paper*)
2. Kinetics of Calcite Nucleation onto Sulfated Chitosan Derivatives and Implications for Water–Polysaccharide Interactions during Crystallization of Sparingly Soluble Salts. B.M. Knight, **R. Mondal**, et al., Crystal Growth & Design, 2024, DOI: [10.1021/acs.cgd.4c00602](https://doi.org/10.1021/acs.cgd.4c00602).
1. Dynamics accelerate the kinetics of ion diffusion through channels: Continuous-time random walk models beyond the mean field approximation. **R. Mondal** and V. Vaissier Welborn, Journal of Chemical Physics, 2024, DOI: [10.1063/5.0188469](https://doi.org/10.1063/5.0188469).

LANGUAGES

- English, Hindi (fluent), Urdu, French, Punjabi, Bengali (professional working proficiency), Korean (limited proficiency).

INTERESTS

- Reviewer for Journal of Open-Source Software, Indian classical violin (Carnatic), history and long-form non-fiction reading.